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The relative importance of neighbours and abiotic environmental conditions for population dynamic parameters of two alpine plant species

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Summary

1 Vegetation removal manipulations over four growing seasons in an alpine *Dryas* heath at Finse, south Norway, were used to examine the role of neighbour vegetation in the population dynamics of the forb *Thalictrum alpinum* and the sedge *Carex vaginata.* We also increased temperature (by open top chambers) and nutrient availability to assess if effects of surrounding vegetation may be modified by environmental change.
2 Removal of neighbour vegetation increased the number of leaves of *Thalictrum* and *Carex*, whereas flowering stems and leaf stalks of *Thalictrum* became significantly shorter. Furthermore, increased plant growth after nutrient addition and warming suggests that both abiotic environmental conditions and biotic interactions limit plant growth at Finse.
3 Our results suggest that there is competition for nutrients at Finse, but that surrounding vegetation may also have facilitative sheltering effects on other species.
4 Interactions between vegetation removal and abiotic factors on population dynamic parameters of *Thalictrum* suggest that plant–plant interactions may be modified if temperature and nutrient availability increase under climate change.
5 The responses of the two target species to the removal of the general neighbour vegetation were less pronounced than their responses to the removal of the dominant *Dryas octopetala* examined in a previous study. This suggests that one dominant species may affect the population dynamics of other species more than the net effect of all the other species in the community in concert.
6 Our results show that biotic interactions may affect alpine plant populations and thereby structure alpine plant communities. Furthermore, the effects of neighbours and abiotic environmental changes on alpine plant growth may depend on the species identity of the neighbours.

Key-words: *Carex vaginata*, climate change, competition, *Dryas* heath, facilitation, nutrient availability, vegetation removal, species interactions, *Thalictrum alpinum*, warming

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Introduction

The relative importance of species interactions and abiotic environmental factors for community structure and processes in areas of contrasting abiotic severity and productivity is unclear (Callaway & Walker 1997; Huston & McBride 2002). Grime (1979) proposed that competition plays an important role in structuring species composition in plant communities where primary production is high. However, the effects of species interactions in areas of low productivity, such as in alpine and arctic plant communities of high abiotic stress, are heavily debated. Under such conditions, competition may have an important role for the population dynamic of individual species because soil resources are in limited supply (e.g. Chapin & Shaver 1985; Tilman 1988; Körner 1999). Alternatively, some ecologists propose that facilitation may have a more important role than competition in environmentally severe habitats because neighbour vegetation may ameliorate the physical environment and thereby

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improve the survival, growth and reproduction of individual plants (e.g. Callaghan & Emanuelson 1985; Bertness & Callaway 1994). It is also possible that interactions (positive or negative) from other species have a low impact on population dynamics compared with direct impacts from abiotic environmental conditions (e.g. Savile 1960; Billings & Mooney 1968; Grime 1977).

Climate change is predicted to have large effects on alpine and arctic plant communities (e.g. Callaghan & Jonasson 1995; Guisan *et al*. 1995; Körner 1995, 1999), and may directly modify abiotic conditions that influence individual plant performance (Arft *et al*. 1999), such as soil moisture and nutrient availability (e.g. Nadelhoffer *et al*. 1991; Harte *et al*. 1995). In combination with the direct warming effect, these factors may have an impact on the internal processes of individual plants, such as growth, reproduction and resource allocation (Arft *et al*. 1999; Dormann & Woodin 2002), with potential cascading effects on the dynamics of populations and the species composition and diversity of plant communities. Moreover, as individual plant species (see for example Henry & Molau 1997) may respond differently to environmental change, global warming may indirectly change the current balance between positive and negative plant–plant interactions and thereby alter dominance hierarchies in plant communities. Thus, a more complete understanding of climate change effects requires that we consider the relative importance of changes in direct abiotic effects vs. indirect biotic effects on important parameters of the population dynamics of species in concert.

The impact of neighbour vegetation on plant community structure, through its impacts on the population dynamic of individual species, may differ spatially and among species (see Hobbie *et al*. 1999; Dormann *et al*. 2004). Moreover, even though interactions from neighbouring vegetation may directly affect community structure, and may be modified by climate change, this effect is largely ignored in models predicting plant community or ecosystem responses to climate change (Mooney 1991; Pacala & Hurtt 1993; Davies *et al*. 1998; Dormann & Woodin 2002). Only a few climate change experiments have focused on species interactions (but see Shevtsova *et al*. 1997; Hobbie *et al*. 1999; Bret-Harte *et al*. 2004; Dormann *et al*. 2004; Klanderud 2005), and most of these have assessed the effects of only one or a few selected species on neighbours (but see Bret-Harte *et al*. 2004). In the natural biota, however, several species may simultaneously have an impact on each other (Bret-Harte *et al*. 2004). Furthermore, climate change may affect all the possible interactions between all the species in a community.

In this study, we first examine the possible effects of removal of all neighbour vegetation on population dynamic parameters of two species (*Thalictrum alpinum* L. and *Carex vaginata* Tausch.) in an alpine plant community. Secondly, we compare results from this study with effects of removal of only the dominant dwarf shrub, *Dryas octopetala* L. (Klanderud 2005), using the same experimental set-up. This will enable us to assess if removal of the dominant species and removal of the general neighbour vegetation have similar or different impacts on the two target species. We use experimental removal of neighbours, warming (open top chambers, OTCs) and nutrient addition to examine the relative role of neighbour vegetation and abiotic environmental conditions on the population dynamics of *Thalictrum alpinum* and *Carex vaginata* and thus to assess whether species interactions may be modified under climate warming. We predict that removal of the whole neighbour vegetation may have greater effect on the target species than removal of only *Dryas* (Klanderud 2005), because numerous species and functional groups in concert are likely to utilize the available resources more completely (e.g. MacArthur 1970; Tilman 1996) than one single species, despite this species being dominant.

Methods

STUDY SYSTEM

The experiment was initiated in early July 2000 and continued until the end of the growing season (late August) in 2003. The study site was situated in a south-west exposed slope of a *Dryas octopetala* heath at *c*. 1500 m elevation on Sandalsnuten, Finse, in the northern part of Hardangervidda (60° N, 7° E) in alpine south-west Norway. The mean summer (June, July, August) temperature at 1222 m elevation at Finse is 6.3 °C (Aune 1993), and mean summer precipitation is 89 mm (Førland 1993). Abundant vascular species in the *Dryas* heath, besides *Dryas octopetala*, are the forbs *Thalictrum alpinum* L., *Potentilla crantzii* Crantz., *Bistorta vivipara* L. and *Cerastium alpinum* L., the dwarf shrub *Salix reticulata* L., the grasses *Festuca vivipara* L. and *Poa alpina* L., and the sedges *Carex vaginata* Tausch., *C. atrofusca* Schkuhr, *C. rupestris* and *Luzula spicata* L. (nomenclature follows Lid & Lid 1994).

EXPERIMENTAL DESIGN

To assess if warming, increased nutrient availability and removal of neighbour vegetation have any effects on components of the population dynamics of *Thalictrum alpinum* and *Carex vaginata*, and if environmental conditions may modify any effect of the surrounding vegetation, we randomly selected 20 plots. We randomly placed OTCs upon 10 of these plots and left the 10 others as controls (ambient).

In each of the 20 main-plots, we selected eight *Thalictrum* and eight *Carex* tillers growing at least 10 cm away from any *Dryas* plant, and inserted half a slow-dissolving NPK-fertilizer stick into the soil *c.* 1 cm upslope of half the individuals of each species immediately after snowmelt and in late July (*c.* 0.2 g N, 0.04 g P and 0.17 g K per tiller/growing season). Thereafter, we clipped the above-ground parts of all neighbour species, and carefully removed below-ground parts that

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could be pulled up with minimal soil disturbance, from a radius of *c.* 5 cm around half of the fertilized tillers and around half of the tillers not receiving any fertilizer. We removed re-growth (mainly bryophytes with minimal below-ground biomass) twice during each of the four growing seasons. This provides a split-plot design (Underwood 2001) with temperature treatment as a fixed factor conducted at main-plot level, 20 plots as a random factor nested within the main-plot factor (temperature), and nutrient addition and removal treatments as fixed factors conducted at subplot level. Thus, within each plot, whether with or without an OTC, two tillers of each of the two target species received each of the possible treatments: nutrient addition, removal of neighbours, nutrient addition and neighbour removal, and control. The responses of these two individuals were averaged prior to statistical analysis.

Experiments with removal of above-ground biomass may be problematic because it leaves roots to decompose, resulting in increased soil nutrients (Putwain & Harper 1970; Berendse 1983). However, these resources have most likely been obtained by competition in the past, and their release may benefit those plants that have previously been denied access to them (Aarssen & Epp 1990). Vegetation removal may also disturb the soil, resulting in a nutrient flush, but here there were probably only minor effects on below-ground processes and soil disturbance.

The OTCs are hexagonally shaped polycarbonate chambers with an inside diagonal of *c.* 1 m, and with qualities as described in Marion *et al*. (1997). OTCs are commonly used in climate change experiments to raise the temperature while minimizing secondary experimental effects, such as changes in atmospheric gas concentrations and ambient precipitation (e.g. Marion *et al*. 1997; Hollister & Webber 2000). The chambers did not affect the duration of snow cover (K. Klanderud, personal observation) and were therefore left in place during winter throughout the experiment. The OTCs increased summer air temperature *c.* 5 cm above ground by *c.* 1.5 °C, and ground temperature by *c.* 2.5 °C. In undisturbed vegetation, the OTCs increased soil temperature *c.* 5 cm below ground by *c.* 1.5 °C (see Klanderud 2005 for further details).

We fenced the site to prevent sheep grazing.

GROWTH MEASUREMENTS

We measured growth and reproductive variables of *Thalictrum alpinum* and *Carex vaginata* in late August after the first (2000), second (2001), third (2002) and reproductive variables only in the fourth (2003) growing season. To assess differences in within-season growth rates, we also measured the growth variables during the second and the third growing season in late May (before any growth had started), late June and late July. However, to enable possible remaining legacy effects caused by the removal treatment to decline, only the 2002 and 2003 measurements were used in the statistical analyses, with the 2000 measurements as covariables.

Sexually reproducing *Carex* tillers normally die the year after flowering, with new tillers growing out from the old (*Carex vaginata*, K. Klanderud and Ø. Totland, personal observation; *C. bigelowii*, see Brooker *et al*. 2001). To simplify interpretation of growth and sexual reproduction of *Carex*, we obtained target tillers at similar developmental stages by selecting flowering *Carex* tillers at the start of the experiment. Thus, we measured the new daughter tillers if the mother tiller died. Daughter tillers may also grow out from living target tillers of both *Thalictrum* and *Carex*, and to obtain an estimate of vegetative growth of the two species, we counted the number of green leaves on each target tiller with daughter tillers if present. The data from the target and daughter tillers are pooled in the statistical analyses. As the number of leaves is not reported per tiller, possible changes may be due either to increased tillering or increased size of the individual tillers. Furthermore, we measured the length and the width of the largest and the smallest leaf and the length of their leaf stalks (*Thalictrum*), and the length of the longest leaf (*Carex*), using a digital caliper. We calculated an approximate leaf area for *Thalictrum* leaves by multiplying the width by the length and used the mean of the values for the largest and smallest leaf to represent average area for that tiller. Number of leaves, average leaf area, and the length of leaf stalks, are parameters commonly used to obtain non-destructive measures of vegetative growth for forbs, whereas number of leaves and the length of the longest leaf are commonly used for sedges (Molau & Edlund 1996; Arft *et al*. 1999). We recorded dead target tillers to estimate mortality. Measuring mortality on clonal plants is, however, not easily applied because individuals usually persist, and only parts of the plants die. The data for *Carex* were not analysed because mortality here coincided with sexual reproduction.

Vegetative regeneration is common for alpine plants. *Thalictrum* and *Carex* grow clonally by producing new tillers from below-ground rhizomes, and seedlings of these species are rarely found on Sandalsnuten. Therefore, our assessment of population dynamic responses relates to effects on mature individuals rather than seedling responses.

SEXUAL REPRODUCTION MEASUREMENTS

To estimate the sexual reproductive effort of *Thalictrum* and *Carex*, we measured the height of flowering stems and collected mature infructescences at the end of each season and recorded the number of flowers and number and weight of dried seeds. Alpine and arctic plants normally do not produce flowers every year (Sonesson & Callaghan 1991), and the same tiller did not reproduce more than once during the experiment for either of the two species. To increase the sample size, we therefore pooled measurements of all flowering tillers from 2002 to 2003 to conduct statistical tests on

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height of flowering stem, number of flowers and seeds, and flowering frequency of *Thalictrum*. There were too few mature seeds to conduct statistical tests on seed weight. In *Carex*, there were not enough flowering tillers to test for any of the sexual reproduction parameters.

STATISTICAL ANALYSES

To examine if warming, nutrient addition and removal of neighbour vegetation had any impact on the growth of *Thalictrum* and *Carex*, and if possible species interactions were affected by the environmental factors, we used general linear models (GLM, SYSTAT 10) with the temperature treatment (main-plot factor), nutrient addition, vegetation removal and their interactions (subplot factors) as fixed factors, and plot nested within temperature as a random factor in a split-plot ANCOVA. We used the first year (2000) measurements as covariables in the analyses to increase the model's ability to detect treatment effects, although this may decrease the ability to detect any responses if these actually occurred during the first year. The effect of temperature was tested over the plot error (main-plot term), whereas all other effects were tested over the model error (subplot term), as described in, for example, Underwood (2001). Analyses were conducted separately for the two species. Data on leaf number, leaf area (*Thalictrum*) and leaf length (*Carex*) were log-transformed to fulfil the ANCOVA assumptions of normality and equal variances. All graphs are shown with untransformed data.

To examine if the removal of neighbour vegetation or the environmental factors had any impact on the growth phenology of *Carex* and *Thalictrum* during the third (2002) growing season, we used repeated-measures ANOVA with the same factors as above, and with tiller identity during three measurements throughout the season (time) as the repeated measures factor. A significant interaction between time and treatment may then indicate that treatments differ in the pattern of growth during the season.

Because of low sample sizes for sexual reproduction, we used Mann–Whitney *U*-tests to assess if the height of flower stems, number of flowers per tiller and number of seeds per flower were affected by the treatments. Analyses were done within each treatment separately (temperature vs. ambient, nutrient addition vs. natural, removal vs. undisturbed). We used Pearson's chi-square to examine if flowering frequency and mortality for *Thalictrum* differed between treatments.

Results

VEGETATIVE GROWTH

Removal of neighbour vegetation increased the number of leaves of *Thalictrum alpinum* by 79.2% (Fig. 1a) and of *Carex vaginata* by 54.5% (Fig. 2a) across all treatments. The *Thalictrum* leaves became 31.3% smaller

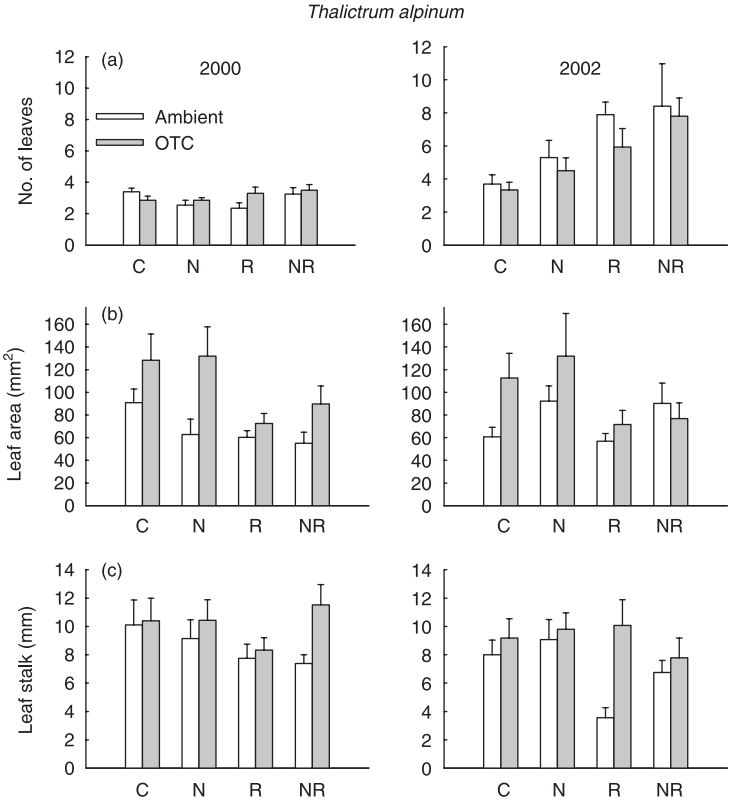


Fig. 1 Means and standard errors of (a) the number of leaves, (b) leaf area, and (c) length of leaf stalks of *Thalictrum alpinum* measured at the start of the experiment in 2000 (ANCOVA pre-treatment values) and by the end of the 2002 growing season in alpine *Dryas* heath vegetation outside (ambient) and inside OTCs, on tillers receiving nutrients (N), removal (R), nutrients and removal (NR), or no treatments (C), at Finse, alpine south Norway. $n = 10$ for each treatment.

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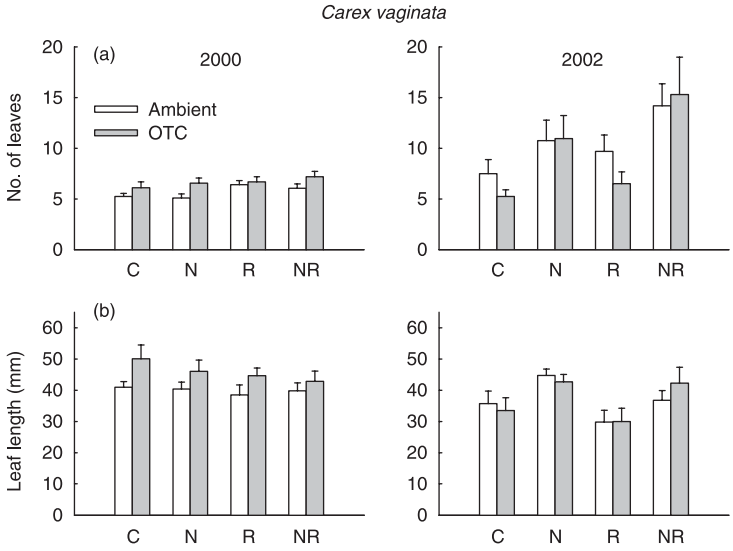


Fig. 2 Means and standard errors of (a) the number of leaves and (b) leaf length of *Carex vaginata* measured in alpine *Dryas* heath vegetation by the start of the experiment in 2000 (ANCOVA pre-treatment values) and by the end of the 2002 growing season outside (ambient) and inside OTCs, on tillers receiving nutrients (N), removal (R), nutrients and removal (NR), or no treatments (C), at Finse, alpine south Norway. $n = 10$ for each treatment.

Table 1 $F_{d.f.}$ and P-values (significant at ≤ 0.05 in bold, < 0.1 in bold italics) of treatments performed in alpine *Dryas* heath vegetation (T = temperature, OTC vs. ambient, main-plot fixed factor; N = nutrient addition, increased vs. ambient, subplot fixed factor; R = removal of vegetation, yes vs. no, subplot fixed factor), their interactions, plot nested within T (random factor), and measurements from 2000 as covariables in a split-plot design ANCOVA on number of leaves of *Thalictrum alpinum* and *Carex vaginata*, average leaf area and length of leaf stalk of the biggest and the smallest leaf of *Thalictrum*, and length of the longest leaf of *Carex*. $n = 10$ for each treatment

	No. of leaves		Leaf area or length		Leaf stalk	
Sources of variation	$F_{d.f.}$	P	$F_{d.f.}$	P	$F_{d.f.}$	P
Thalictrum alpinum						
T	$0.60_{1,18}$	0.448	$0.15_{1,18}$	0.705	$\mathbf{7.39_{1,18}}$	**0.014**
N	$1.04_{1,46}$	0.313	$\mathbf{5.95}_{1,46}$	**0.018**	$1.95_{1,46}$	0.169
R	$\mathbf{17.75}_{1,46}$	**< 0.001**	$0.04_{1,46}$	0.841	$\mathbf{7.71}_{1,46}$	**0.008**
T × N	$0.71_{1,46}$	0.402	$\mathbf{6.83}_{1,46}$	**0.012**	$\mathbf{6.44}_{1,46}$	**0.014**
T × R	$0.00_{1,46}$	0.966	$0.09_{1,46}$	0.767	$\mathbf{5.44}_{1,46}$	**0.024**
N × R	$1.38_{1,46}$	0.246	$0.92_{1,46}$	0.342	$0.03_{1,46}$	0.864
T × N × R	$1.78_{1,46}$	0.189	$0.55_{1,46}$	0.462	$\mathbf{7.45}_{1,46}$	**0.009**
Plot (T)	$1.12_{1,46}$	0.360	$\mathbf{2.28}_{1,46}$	**0.011**	$1.28_{1,46}$	0.243
2000 measure	$0.08_{1,46}$	0.773	$\mathbf{18.99}_{1,46}$	**< 0.001**	$1.58_{1,46}$	0.215
Carex vaginata						
T	$0.84_{1,18}$	0.373	$0.15_{1,18}$	0.704		
N	$\mathbf{22.98}_{1,44}$	**< 0.001**	$\mathbf{13.67}_{1,43}$	**0.001**		
R	$\mathbf{5.05}_{1,44}$	**0.030**	$\mathbf{\mathit{2.94}}_{1,43}$	***0.094***		
T × N	$\mathbf{3.97}_{1,44}$	**0.053**	$0.25_{1,43}$	0.620		
T × R	$0.06_{1,44}$	0.812	$0.90_{1,43}$	0.347		
N × R	$0.23_{1,44}$	0.633	$0.003_{1,43}$	0.959		
T × N × R	$2.38_{1,44}$	0.130	$0.006_{1,43}$	0.938		
Plot (T)	$\mathbf{\mathit{1.78}}_{1,44}$	***0.060***	$1.30_{1,43}$	0.234		
2000 measure	$0.21_{1,44}$	0.647	$0.69_{1,43}$	0.410		

and the *Carex* leaves became 10.2% shorter after vegetation removal, although this effect was not significant for *Thalictrum* and only close to significant for *Carex* (Table 1, Figs 1b and 2b). Furthermore, vegetation removal decreased the length of *Thalictrum* leaf stalks by 22.6%, although a significant three-way interaction indicated that temperature, nutrient and removal treatments modified each other's impacts on the leaf stalk lengths (Table 1, Fig. 1c). Two-way ANCOVAs performed on tillers inside and outside OTCs separately, suggested that the removal treatment significantly decreased the length of *Thalictrum* leaf stalks outside the OTCs ($F_{1,34} = 10.07$, $P = 0.003$) but not inside ($F_{1,32} = 0.12$, $P = 0.728$) (Fig. 1c). Moreover, close to significant and significant interactions between nutrient addition and vegetation removal outside ($F_{1,34} = 3.87$, $P = 0.057$)

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and inside ($F_{1,32} = 5.15$, $P = 0.03$) the OTCs (Fig. 1c), suggested that the responses to the combined effect of these factors were less than the sum of the separate effects. Thus, warming increased the length of *Thalictrum* leaf stalks, but only in combination with added nutrients or after vegetation removal (Table 1, Fig. 1c).

Warming or nutrient addition had no effects on the number of *Thalictrum* leaves (Table 1, Fig. 1a). For *Carex*, on the other hand, an interaction between warming and nutrient addition suggests that the responses to the combined effect of these factors were less than the sum of the separate effects (Table 1, Fig. 2a). Overall, warming and nutrient addition increased leaf area of *Thalictrum* by 43.5% and 30.3%, respectively. The warming effect had already occurred by the end of the first growing season (significant 2000 measure, Table 1, Fig. 1b), which may explain why no such response was detected by the ANCOVA. An interaction between warming and nutrient addition suggests that the combined effect of these treatments was larger than the effect of each treatment alone (Table 1, Fig. 1b). Nutrient addition increased the length of *Carex* leaves by 28.8%, and warming had no effect on the length of *Carex* leaves (Table 1, Fig. 2b).

There were no significant treatment effects on growth phenology of *Thalictrum* or *Carex* (non-significant interactions between time and treatment in the repeated measures ANOVA).

SEXUAL REPRODUCTION AND MORTALITY

For *Thalictrum*, 17.5%, 18.8%, 46.3% and 10% of the tillers reproduced sexually in 2000, 2001, 2002 and 2003, respectively. Removal of neighbours or nutrient addition had no effect on its flowering frequency (Pearson's $\chi^2 < 0.001$, $P > 0.10$ in both cases). The height of the flower stems of *Thalictrum* were increased by 55.8% by warming (Mann–Whitney $U = 198$, $P = 0.002$) and reduced, across all treatments, by 22.4% following neighbour removal (Mann–Whitney $U = 165$, $P = 0.039$, Fig. 3a). None of the treatments had any significant effects on the number of flowers and seeds (Table 2, Fig. 3b,c), although the combination of warming and nutrient addition appeared to increase the number of flowers per tiller (Fig. 3b).

For *Carex*, 98.8%, 0%, 5% and 8.8% of the target tillers reproduced sexually in 2000, 2001, 2002 and 2003, respectively.

None of the treatments had any significant effects on *Thalictrum* mortality (Pearson's $\chi^2 < 0.001$, $P > 0.98$ in all cases).

Discussion

Neighbour vegetation clearly affected plant growth and, potentially, population dynamics of *Thalictrum alpinum* and *Carex vaginata* at alpine Finse. The responses of the two species to the removal of neighbours in this study were, however, not as pronounced as their responses to the removal of the dominant dwarf shrub *Dryas octopetala* in the same system (Klanderud 2005). This result may suggest that the role of one dominant species in affecting population dynamics of plant species may be more important than the net effect of all the other species in this community.

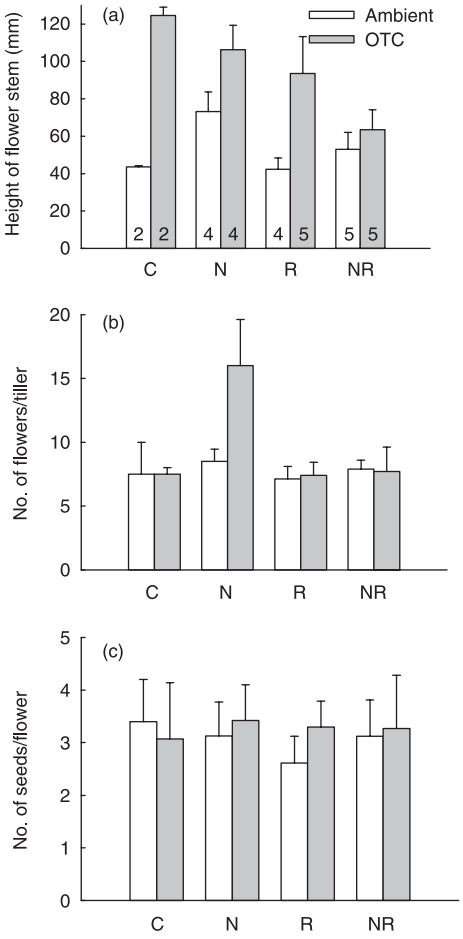


Fig. 3 Means and standard errors of (a) the height of flower stems, (b) number of flowers tiller^{-1}, and (c) number of seeds flower^{-1} of *Thalictrum alpinum* measured in alpine *Dryas* heath vegetation by the end of the 2002 and 2003 growing season outside (ambient) and inside OTCs, on tillers receiving nutrients (N), removal (R), nutrients and removal (NR), or no treatment (C), at Finse, alpine south Norway. Numbers in the columns in (a) are sample sizes for each treatment in (a), (b) and (c).

EFFECTS OF VEGETATION REMOVAL

Flower stems and leaf stalks of *Thalictrum*, and leaves of *Carex*, became shorter (although only marginally significant for *Carex*), whereas leaf number of *Thalictrum* and *Carex* increased after removal of neighbour vegetation (Table 2). These opposing responses, which occur in both species, are in line with the responses to *Dryas* removal (Klanderud 2005), and suggest that individual tillers change their internal resource allocation after vegetation removal. The results are, however, in contrast to removal treatments in other arctic sites, where there have been few detected effects of neighbours

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Table 2 Summary of treatment responses with significant effects ($P < 0.05$, $P < 0.1 > 0.05$ in parentheses) on growth and reproductive parameters of *Thalictrum alpinum* and *Carex vaginata* growing in alpine *Dryas* heaths, as tested by different methods (see text). + = positive response; – = negative response; empty cells indicate no response ($P \geq 0.1$); and dots (.) indicate that no tests are done. Treatment abbreviations as in Table 2

Species and parameters	Treatments and responses						
	+T	+ N	+ R	+ T + N	+ N + R	+ T + R	+ T + N + R
Thalictrum alpinum							
No. of leaves			+				
Leaf area				+			
Length of leaf stalk		–	–	+	(–)		–
Height of flower stem	+		–	.	.	.	.
No. of flowers				.	.	.	.
No. of seeds				.	.	.	.
Flowering frequency					.	.	.
Mortality				.	.	.	.
Carex vaginata							
No. of leaves		+	+	+			
Leaf length		+	(–)				

(Jonasson 1992; Hobbie *et al*. 1999; Bret-Harte *et al*. 2004), and thus no evidence for species interactions (but see Dormann *et al*. 2004). The increased number of leaves of *Thalictrum* and *Carex* after vegetation removal suggests competitive rather than facilitative impacts of other species at Finse, which is in line with Klanderud (2005) and some other removal experiments in alpine or subarctic sites (Aarssen & Epp 1990; Shevtsova *et al*. 1995). However, plants can be suppressed by some neighbours, and at the same time, benefit from the presence of others (Aarssen & Epp 1990; Dormann *et al*. 2004), and the shorter leaf stalks, flower stems and leaves after removal treatment may be caused by a loss of facilitative shelter against low temperature and strong winds. The positive effects of warming on flower stems and leaf stalks of *Thalictrum* further suggests a possible benefit from protective neighbours, which is regarded as an important factor in extreme environments (e.g. Carlsson & Callaghan 1991; Callaway *et al*. 2002). Bret-Harte *et al*. (2004) found possible facilitation effects from one shrub (*Ledum palustre*) on forb species (*Pedicularis* spp.) in the Alaskan tundra, and Shevtsova *et al*. (1997) found negative effects on *Vaccinium vitis-idaea* after removal of *Empetrum nigrum*. Moreover, Shevtsova *et al*. (1995) found decreased shoot length in various species after neighbour removal, which they suggested to be due to either a loss of facilitative protection or a release from negative shading effects. The latter explanation assumes that plants may produce larger leaves (e.g. Dormann & Woodin 2002; Totland & Esaete 2002) or longer shoots to compensate for shading and to position flowers, seeds or photosynthetically active tissue above neighbour plants (Grime 1979). However, the positive growth responses to warming, and the negative effect of removal on leaf stalks outside but not inside the OTCs (Fig. 1c), suggest that, at Finse, at least part of the effect is due to facilitation. Furthermore, a larger relative decrease of *Carex* leaf lengths after *Dryas* removal (Klanderud 2005) than after removal of other species (this study), may suggest a greater facilitative effect of *Dryas* than of other species on the vegetative growth of *Carex*.

EFFECTS OF CLIMATE CHANGE SIMULATION

Both warming and nutrient availability limited the growth of *Thalictrum* and *Carex* at Finse, with the two factors often showing synergistic effects. Warming alone had only minor effects on vegetative growth of either species, but increased the height of flowering stems of *Thalictrum* (Table 2). Nutrient addition on the other hand, had only minor effects on the sexual reproduction of *Thalictrum*, but increased the vegetative growth of both species (Table 2). Other climate change experiments have also found that nutrients limit plant growth in alpine and arctic environments and that enhanced temperature alone has only minor effects (Dormann & Woodin 2002). Often, however, studies have found synergistic effects of warming and nutrients together, suggesting that the conditions of one factor may affect responses to another factor (e.g. Chapin & Shaver 1985; Robinson *et al*. 1998; Klanderud & Totland, in press).

EFFECTS OF WHOLE VEGETATION VERSUS ONE DOMINANT SPECIES

Although the vegetation removal in this study had generally similar effects on the population dynamic parameters of *Thalictrum alpinum* and *Carex vaginata* as the removal of the dominant *Dryas octopetala* had in a previous study by Klanderud (2005), the overall effects of *Dryas* appeared to be larger. For example, vegetation removal (this study) had no significant effects on the combination of seasonal growth and effect sizes of *Thalictrum* and *Carex*, whereas *Dryas* removal did have significant effects (Klanderud 2005). Furthermore, nutrient addition increased leaf sizes of *Thalictrum* and *Carex* tillers surrounded by general vegetation (this study), but had no effect on leaf sizes of tillers surrounded by

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Dryas (Klanderud 2005). Moreover, flowering frequency of *Thalictrum* was considerably higher when surrounded by general vegetation than inside *Dryas* mats. All these results may suggest that *Dryas* controls more of the population dynamics of the two target species than the sum of all other species in the community in concert. Moreover, the role of *Dryas*, both as a facilitator and a competitor, appears to be greater than the net effect of the general vegetation. Although individual species may have imposed equally large effects on the target species as did *Dryas*, the effects of species acting in opposite directions could have cancelled this out, leading to a smaller net effect compared with that of *Dryas*. Moreover, the larger *Dryas* effect may have been due to the *Dryas* biomass removed in the previous experiment being more than the biomass of surrounding vegetation removed here. Indeed, it may be important to distinguish between the roles of the amount vs. the identity of the biomass removed to understand ecosystem functioning (Díaz *et al*. 2003). Unfortunately, our data are not suitable for testing these potentially opposing effects.

Our results are in contrast to our predictions, where we anticipated a larger effect of vegetation removal because numerous species and functional groups are expected to utilize more resource niches than only one species (e.g. Tilman 1996). The weaker effects of vegetation removal than *Dryas* removal are, however, partly in line with Bret-Harte *et al*. (2004), who did not find any effects of neighbour removal in the Alaskan tussock tundra, even though nutrient release increased by up to two orders of magnitude. They suggested that rigid niche differentiation and complementarities of nutrient uptake of tundra plants make species unable to utilize additional resources released by neighbour removal (Bret-Harte *et al*. 2004). Nevertheless, *Dryas*, which already dominated the resource use in our site, apparently strengthened its dominance position when resources increased after nutrient addition. Moreover, warming decreased flowering frequency and nutrient addition decreased seed number of *Thalictrum* tillers associated with *Dryas* (Klanderud 2005), whereas these treatments had no effects on individuals surrounded by other species (this study). This may suggest that possible negative effects of *Dryas* may be larger when resources increase, most likely because *Dryas* benefits more from increased temperature and nutrient availability than the majority of other species.

EFFECTS OF ENVIRONMENTAL CONDITIONS ON SPECIES INTERACTIONS

Warming and nutrient addition significantly influenced the effect of vegetation removal on population dynamic parameters of *Thalictrum*, suggesting that climate change may modify species interactions at alpine Finse. In particular, the negative effect of removal on *Thalictrum* leaf stalks outside the OTCs but not inside, may suggest a benefit of protective neighbours that could become redundant under global warming. Similarly, the interaction between nutrient addition and vegetation removal on *Thalictrum* leaf stalks may suggest that, after removal of potential competitors, there is no benefit of the added nutrients, i.e. the plants are no longer nutrient limited.

The results of this study, combined with those of Klanderud (2005), highlight the role of both positive (e.g. Callaghan & Emanuelson 1985; Bertness & Callaway 1994) and negative (e.g. Chapin & Shaver 1985; Tilman 1988; Körner 1999; Dormann *et al*. 2004) species interactions in structuring alpine plant communities of high abiotic stress. Global warming may cause a shift towards an increased role of competition in cold environments, because the role of facilitation appears to decrease under warming (this study; Callaway *et al*. 2002). Moreover, increased nutrient availability, due to higher decomposition and mobilization of resources in warmer soils, may change competition hierarchies, resulting in an increased role of competition for light and space. Significant interactions between removal manipulations and climatic factors in this study and that of Klanderud (2005) highlight the importance of understanding how climate change modifies species interactions, in order to predict future plant community responses to global warming with greater precision.

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